

PCRF Transformer Reliability Executive Report

Causal Reliability Flow & Derivative-Weighted Diagnostics Dashboard for QWEN/QWEN2.5-0.5B

1. Executive Summary & Core Gating Status

- **Target Architecture Profile:** QWEN/QWEN2.5-0.5B
- **Automated Path C Gating:** DISABLED (Discrete Exact Match accuracy utilized)
- **Composite PCRF Adoption Index:** 44.39 / 100
- **Strategic Deployment Directive:** REJECT ADOPTION: RESTORE BASELINE CONFIGURATIONS
- **Production Security Risk Level:** ● [HIGH OVER-STEERING RISK | BLOCK PARAMETER MUTATIONS]

System Reliability Metrics At-a-Glance

Performance Parameter	Baseline (Pre-Intervention)	Post-Regularization	Net Gain / Loss Delta
Seen Validation Accuracy	60.00%	60.00%	+0.00%
Unseen Generalization Acc	65.00%	60.00%	-5.00%
Seen Validation Loss (NLL)	4.54374	4.13632	-0.40742
Unseen Validation Loss (NLL)	4.26385	4.04191	-0.22193
Unseen Perplexity (PPL)	71.08	56.94	-14.15

2. Integrated PCRF Scoreboard

Feature Track / Module	Baseline Value	PCRF Result Value	Track Score	Gating Status
Derivatives	0.00 (Unmeasured)	0.00372 (Avg Sensitivity)	3.7/100	SAFE_TO_APPLY
Curriculum Curation	Uniform Selection (Std=0.0)	PCRF Prioritized (Std=3.78)	75.6/100	SAFE_TO_APPLY
Structural Depth Monitor	Unmonitored Depth	Chain Reliability: 35.11%	35.1/100	SAFE_TO_APPLY
Safe SFT Regularization	Unseen Acc: 65.0%	Unseen Acc: 60.0%	60.0/100	DO_NOT_APPLY

3. Deep-Dive Diagnostics: Structural Bottlenecks & Representation Decay

Dynamic Decay Curve Insights for QWEN/QWEN2.5-0.5B

Causal analysis on QWEN/QWEN2.5-0.5B revealed the following trends across its structural segments:

- **Calculated System Chain Reliability (R_{sys}):** 35.1096%
- **Triggered Structural Flaw Heuristics:** None

Layer-wise Survival & Birnbaum Sensitivity Index

Layer	Survival Probability (r_l)	Birnbaum Derivative (D_R)
Block 00	88.87%	0.39507
Block 01	91.44%	0.38395
Block 02	97.85%	0.35879
Block 03	99.17%	0.35402
Block 04	99.03%	0.35453
Block 05	99.02%	0.35459
Block 06	98.92%	0.35492
Block 07	98.73%	0.35560
Block 08	98.73%	0.35560
Block 09	98.81%	0.35532
Block 10	98.76%	0.35549
Block 11	98.84%	0.35521
Block 12	98.84%	0.35521
Block 13	98.83%	0.35526
Block 14	98.64%	0.35594
Block 15	98.64%	0.35594
Block 16	98.47%	0.35657
Block 17	98.21%	0.35748
Block 18	97.92%	0.35857
Block 19	97.42%	0.36040
Block 20	96.61%	0.36340
Block 21	86.46%	0.40606
Block 22	85.08%	0.41265
Block 23	78.33%	0.44822

- **Dynamic Layer Integrity Trend Analysis:** U-Shaped Bottleneck Pattern (decay at boundaries)

- Input Segment Average (First 15%): 92.72%
- Highway Segment Average (Middle 70%): 98.53%
- Output Segment Average (Last 15%): 83.29%

4. Empirical Perturbation Analysis: Layer-wise Sensitivity

- **Average System Sensitivity to Perturbation:** 0.00418
- **Highest Sensitivity Bottleneck Layer:** Layer 4 (Max Drift Delta: 0.01112)

Layer Index	Clean Target Prob	Perturbed Target Prob	Delta Prob
Layer 0	9.70%	9.49%	0.00204
Layer 1	9.70%	8.81%	0.00883
Layer 2	9.70%	8.85%	0.00848
Layer 3	9.70%	9.28%	0.00418
Layer 4	9.70%	8.58%	0.01112
Layer 5	9.70%	10.21%	-0.00518
Layer 6	9.70%	10.70%	-0.01005
Layer 7	9.70%	10.28%	-0.00584
Layer 8	9.70%	10.29%	-0.00596
Layer 9	9.70%	9.87%	-0.00172
Layer 10	9.70%	9.93%	-0.00229
Layer 11	9.70%	9.85%	-0.00151
Layer 12	9.70%	10.55%	-0.00855
Layer 13	9.70%	9.53%	0.00170
Layer 14	9.70%	9.69%	0.00010
Layer 15	9.70%	9.64%	0.00054
Layer 16	9.70%	9.23%	0.00462
Layer 17	9.70%	9.31%	0.00387
Layer 18	9.70%	9.32%	0.00377
Layer 19	9.70%	9.31%	0.00388
Layer 20	9.70%	9.31%	0.00388
Layer 21	9.70%	9.62%	0.00075
Layer 22	9.70%	9.65%	0.00044
Layer 23	9.70%	9.59%	0.00110

5. Curriculum Prioritization & Semantic Error Concentration

The curriculum prioritization scoring system matches training prompts with a priority score:

$$\text{Priority Score} = \text{NLL} \times \left(1.0 + \sum_l \Delta_l \right)$$

This mathematical approach isolates low-impact prompts and prioritizes abstract syntax and logical structures. For example, QWEN/QWEN2.5-0.5B consistently struggles with advanced computer science and coding syntax prompts (high priority scores), while basic factual lookups (like capitals) remain stable.

Top 5 Highest Cascade-Risk Training Prompts

ID	Prompt	Target Completion	Priority Score
79	The reserved programming keyword used to initiate routine blocks in Python is	def	18.42
43	The element with atomic number 6 is	Carbon	18.34
78	The reserved programming keyword used to declare structural blueprints in Python is	class	16.87
66	The digital counting framework representing information with 0 and 1 is	Binary	16.68
63	To store keyed associative records with rapid O(1) lookup, developers choose a	Map	14.99

Bottom 5 Lowest Cascade-Risk Training Prompts

ID	Prompt	Target Completion	Priority Score
19	The official capital city of Turkey is	Ankara	4.27
40	The official capital city of Indonesia is	Jakarta	4.23
29	The official capital city of Ireland is	Dublin	4.04
5	The official capital city of Japan is	Tokyo	3.93
16	The official capital city of Argentina is	Buenos Aires	3.56

6. Root-Cause Debugging Hub: Failed Generations

The following table records the precise prompts where the model failed to generate an exact semantic match during validation evaluation. Use this trace to identify and rectify model logic regressions.

Split	ID	Prompt	Expected Target	Baseline Output	Candidate Output	Baseline NLL
seen_val	86	<i>The noble element designated by atomic number 10 is</i>	Neon	` ____`		
A. Sodium`	` ____`					
A. Sodium`	10.9375					
seen_val	87	<i>The volatile element designated by atomic number 16 is</i>	Sulfur	` ____`		
A. Fluor`	` ____`					
A. Fluor`	3.4583					
seen_val	89	<i>The yellow dwarf star supporting life at the center of our solar system is the</i>	Sun	most distant star in the Milky	closest to Earth, and it	3.2344
seen_val	90	<i>Mechanical acoustics are completely incapable of moving across a spatial</i>	Vacuum	domain. The only way to	medium. They are not able	13.5625
seen_val	96	<i>To enforce unique constraints with no duplicated items, algorithms utilize a</i>	Set	technique called "hashing."	technique called "hashing."	7.0625
seen_val	98	<i>An execution failure originating from incorrect program logic is called a</i>	Bug	` ____` failure.		
A. Logical`	` ____` failure.					
A. Logical`	11.1875					
seen_val	99	<i>A standardized text notation representing</i>	JSON	used in the field of structural	used in the field of structural	9.5625

Split	ID	Prompt	Expected Target	Baseline Output	Candidate Output	Baseline NLL
seen_val	100	<i>complex structural records is</i>				
		<i>The active keyword used to bind external packages into Python script scopes is</i>	<i>import</i>	<i>`__</i>		
		A. from`		<i>`__</i>		
A. from`	7.3438					
unseen_val	106	<i>Mammalian red blood cells are chemically responsible for transporting vital</i>	<i>Oxygen</i>	<i>substances in the body. Which</i>	<i>substances in the body. Which</i>	10.0000
unseen_val	109	<i>The dual-helix macromolecule housing core genetic blueprints is</i>	<i>DNA</i>	<i>`the</i>		
		A. DNA`		<i>`the</i>		
A. nucleoid`	6.2188					
...	...	...	...	...	...	(And 6 more trace details)

7. Actionable 4-Phase Enterprise Upgrade Strategy for QWEN/QWEN2.5-0.5B

To transform these automated insights into performance upgrades, the following engineering map is recommended for deployment:

Phase 1: Selective SFT Boundary Layer Regularization

Stop fine-tuning all model weights. Since intermediate layers are naturally self-correcting, freeze them. Concentrate training compute budgets exclusively on the high-risk boundary layers (Layers 0–1 and late projection blocks) to shrink training memory footprints by up to 50% while protecting historical parameter arrays from catastrophic forgetting.

Phase 2: Automated PCRF Dataset Curation

Deploy the `curriculum_scores.csv` output as a filtration gate before SFT training runs. Discard the bottom 30% to 50% of low-priority, high-redundancy training samples, saving massive amounts of GPU compute hours while ensuring the optimization loop is focused entirely on high-impact training samples.

Phase 3: Live Production "Canary" Monitors

Deploy the continuous Structural PCRF Plugin as an active endpoint wrapper. By feeding a tiny perturbed variant alongside the main user prompt, you can calculate the system chain reliability (R_{sys}) in real-time. If R_{sys} collapses below 75% for a given user query, route that query to a safe fallback model before a hallucinated or incorrect output reaches the user.

Phase 4: Automated CI/CD Governance Safety Gates

Integrate the Safe PCRF Controller as a hard gate in your Git-Ops pipelines. No custom-trained model should ever be promoted to your server registry unless it satisfies the continuous Path C non-inferiority constraints (protecting seen validation losses while ensuring a relative generalization gain of at least 5% on unseen splits).

8. Compute Environment & Host Profile Audit

- **Host Platform:** Linux 6.1.0-49-cloud-amd64
- **Detected CPU Cores:** 8
- **Host Memory Capacity:** 29.38 GB
- **GPU Platform Context:** Tesla T4 (14.56 GB VRAM)

Report programmatically generated by PCRF Reliability Suite v1.